

第十七章 含参变量的积分.

例. 设 $f(x, a) = \begin{cases} -\frac{2}{a^2}(x-a), & 0 \leq x \leq a \\ 0 & a < x \leq 1. \end{cases} \quad a \in (0, 1).$

$\forall a > 0: f(x, a) \in C[0, 1].$

$f(x, a) \in C([0, 1] \times (0, 1))$

$a \rightarrow 0:$

$a=0$ 时, 定义

$f(x, 0) \equiv 0, \quad x \in (0, 1]$

$= \lim_{a \rightarrow 0} f(x, a) \equiv 0$

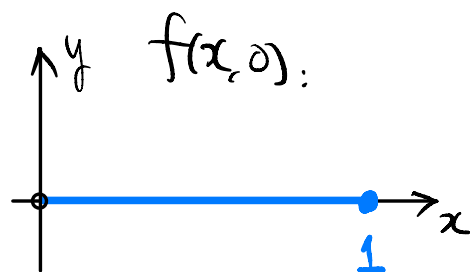
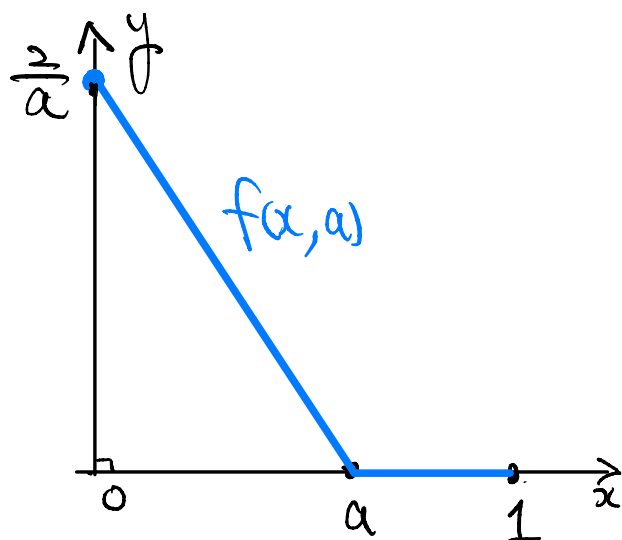
$\int_0^1 f(x, a) dx = \frac{1}{2} a \cdot \frac{2}{a} = 1;$

$\int_0^1 f(x, 0) dx = 0;$

$\lim_{a \rightarrow 0} \int_0^1 f(x, a) dx = 1 \neq \int_0^1 \lim_{a \rightarrow 0} f(x, a) dx = 0.$

极限与积分. 顺序不能交换.

求极限不能“穿过积分”符号!



定理1. (含参量积分的连续性) 令 $D = [a, b] \times [\alpha, \beta]$.

$f(x, y) \in C(D)$. $\forall x \in [a, b]$: $f(x, y) \in C[\alpha, \beta]$, 则

$$I(x) = \int_{\alpha}^{\beta} f(x, y) dy \in C[a, b]$$

即 $\forall x_0 \in [a, b]$: $x \rightarrow x_0$: $I(x) \rightarrow \underline{I(x_0)}$.

$$\lim_{x \rightarrow x_0} \int_{\alpha}^{\beta} f(x, y) dy = \int_{\alpha}^{\beta} \lim_{x \rightarrow x_0} f(x, y) dy$$

证明: 设 $x_0 \in [a, b]$ 固定. 要证: $I(x)$ 在 x_0 连续.

$$\begin{aligned} I(x) - I(x_0) &= \int_{\alpha}^{\beta} f(x, y) dy - \int_{\alpha}^{\beta} f(x_0, y) dy \\ &= \int_{\alpha}^{\beta} (f(x, y) - f(x_0, y)) dy \end{aligned}$$

$$|I(x) - I(x_0)| \leq \int_{\alpha}^{\beta} |f(x, y) - f(x_0, y)| dy$$

$\because f(x, y) \in C(D)$: D 是有界闭集.

$\therefore f(x, y)$ 在 D 上一致连续.

$\forall \varepsilon > 0$. $\exists \delta > 0$: $\forall (x_1, y_1), (x_2, y_2) \in D$: 只要

$$|x_1 - x_2| < \delta \quad |y_1 - y_2| < \delta.$$

$$\text{就有} \quad |f(x_1, y_1) - f(x_2, y_2)| < \varepsilon / (\beta - \alpha).$$

$\forall y \in [\alpha, \beta]$: 取 $y_1 = y_2 = y$, $x_2 = x_0$: 则 $\forall x$: 只要

$$|x - x_0| < \delta: \text{就有} \quad |f(x, y) - f(x_0, y)| < \frac{\varepsilon}{\beta - \alpha} \quad (\forall y \in [\alpha, \beta])$$

$$\Rightarrow |I(x) - I(x_0)| \leq \int_{\alpha}^{\beta} \underbrace{|f(x, y) - f(x_0, y)|}_{< \frac{\varepsilon}{\beta - \alpha}, \forall y \in [\alpha, \beta]} dy$$

$$< \frac{\varepsilon}{\beta - \alpha} \cdot (\beta - \alpha) = \varepsilon$$

$$\therefore \lim_{x \rightarrow x_0} I(x) = I(x_0).$$

$$\text{即 } \lim_{x \rightarrow x_0} \int_{\alpha}^{\beta} f(x, y) dy = \int_{\alpha}^{\beta} f(x_0, y) dy = \int_{\alpha}^{\beta} \lim_{x \rightarrow x_0} f(x, y) dy$$

交换次序

注: 若 $f(x, y) \in C(D) = C([a, b] \times [\alpha, \beta])$ 则

$$I(x, y) = \int_{\alpha}^y f(x, t) dt \in C([a, b] \times [\alpha, \beta]).$$

(Ex.)

定理 2. (求导与积分次序可交换性)

设 $D = [a, b] \times [\alpha, \beta]$, $f(x, y) \in C(D)$, $\frac{\partial f(x, y)}{\partial x} \in C(D)$,

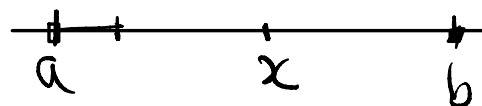
$$\text{则 } I(x) = \int_{\alpha}^{\beta} f(x, y) dy \in C^1[a, b],$$

$$I'(x) = \frac{d}{dx} I(x) = \int_{\alpha}^{\beta} \frac{\partial}{\partial x} f(x, y) dy$$

证明: $\forall x \in [a, b], \Delta x \neq 0, x + \Delta x \in [a, b]$.

$$I(x) = \int_{\alpha}^{\beta} f(x, y) dy$$

$$I(x + \Delta x) = \int_{\alpha}^{\beta} f(x + \Delta x, y) dy.$$



$$\Delta I = I(x+\Delta x) - I(x) = \int_a^b (f(x+\Delta x, y) - f(x, y)) dy$$

$$\frac{\Delta I}{\Delta x} = \int_a^b \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} dy$$

“补充定义”

$I(x+\Delta x)$ 关于 Δx 连续; $f(x, y)$ 关于 Δx 连续.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta I}{\Delta x} = \int_a^b \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} dy$$

$$= \int_a^b \frac{\partial f(x, y)}{\partial x} dy$$

或者如下:

$\therefore f(x, y)$ 关于 x 有偏导数:

$$f(x+\Delta x, y) - f(x, y) = f_x(x+\theta\Delta x, y) \Delta x$$

$$\therefore \frac{\Delta I}{\Delta x} = \int_a^b f_x(x+\theta\Delta x, y) dy \quad \exists 0 < \theta < 1.$$

$\therefore f_x(x, y) \in C(D)$:

$\therefore \Delta x \rightarrow 0: f_x(x+\theta\Delta x, y) \rightarrow f_x(x, y)$ —— 关于 Δx 连续.

$$\therefore \lim_{\Delta x \rightarrow 0} \frac{\Delta I}{\Delta x} = \int_a^b \lim_{\Delta x \rightarrow 0} f_x(x+\theta\Delta x, y) dy$$

$$= \int_a^b f_x(x, y) dy$$

$$\therefore I'(x) = \int_a^b f_x(x, y) dy$$

$$I(x) = \int_a^x f(y) dy : \quad \forall f \in C[a, b], \quad \forall x$$

$$F'(x) = f(x) : \Rightarrow F(b) - F(a) = \int_a^b f(y) dy$$

$$F(x) - F(a) = \int_a^x f(y) dy.$$

$$F'(x) = \left(\int_a^x f(y) dy \right)' = f(x)$$

$$\Rightarrow g(x) \triangleq \int_a^x f(y) dy : \quad \forall x \in [a, b], \quad g'(x) = f(x)$$

$$x = \varphi(t) \quad \forall t \in [a, b] \Rightarrow g(\varphi(t)) = \int_a^{\varphi(t)} f(y) dy$$

$$\frac{d}{dt} \left(\int_a^{\varphi(t)} f(y) dy \right) = \frac{d}{dt} g(\varphi(t))$$

$$= g'(\varphi(t)) \cdot \varphi'(t)$$

$$= \underbrace{f(\varphi(t)) \cdot \varphi'(t)}_{\updownarrow}$$

$$\left(\int_a^{\varphi(t)} f(y) dy \right)' = f(\varphi(t)) \cdot \varphi'(t)$$

定理 3. (含参量变上下限积分的求导公式)

设 $D = [a, b] \times [\alpha, \beta]$, $f(x, y) \in C(D)$, $\varphi(x), \psi(x) \in C[a, b]$,
 $\alpha \leq \varphi(x), \psi(x) \leq \beta, \quad \forall x \in [a, b]. \quad \forall x$

$$J(x) = \int_{\varphi(x)}^{\psi(x)} f(x, y) dy \in C[a, b].$$

$$\begin{aligned}
 \text{证明: } J(x) &= \int_{\varphi(x)}^a f(x, y) dy + \int_a^{\psi(x)} f(x, y) dy \\
 &= \int_a^{\psi(x)} f(x, y) dy - \int_a^{\varphi(x)} f(x, y) dy \\
 &= g(\psi(x)) - g(\varphi(x))
 \end{aligned}$$

$$\text{其中 } g(u) \triangleq \int_a^u f(x, y) dy: \quad \text{关于 } u \text{ 连续};$$

$$u = \varphi(x), \psi = \psi(x) \text{ 连续} \Rightarrow g(\varphi(x)), g(\psi(x))$$

$$\text{在 } [a, b] \text{ 上连续.}$$

$$\Rightarrow J(x) \in C[a, b]$$

定理4. (含参量变上下限积分的求导公式)

$$\text{设 } D = [a, b] \times [\alpha, \beta], f(x, y) \in C(D), \frac{\partial f}{\partial x} \in C(D).$$

$$\varphi(x), \psi(x) \in C^1[a, b], \quad \text{且 } \forall x \in [a, b], \alpha \leq \varphi(x), \psi(x) \leq \beta,$$

$$\text{则 } J(x) = \int_{\varphi(x)}^{\psi(x)} f(x, y) dy \in C^1[a, b].$$

$$\begin{aligned}
 \text{证: } J'(x) &= \int_{\varphi(x)}^{\psi(x)} \frac{\partial f}{\partial x}(x, y) dy + f(x, \psi(x)) \cdot \psi'(x) \\
 &\quad - f(x, \varphi(x)) \cdot \varphi'(x).
 \end{aligned}$$

证明: 令 $F(x, u, v) \triangleq \int_v^u f(x, y) dy$ $u, v \in [\alpha, \beta]$

$$J(x) = \int_{\varphi(x)}^{\psi(x)} f(x, y) dy = F(x, \psi(x), \varphi(x))$$

由 $F(x, u, v)$ 与 $u = \psi(x)$, $v = \varphi(x)$ 复合而得.

$\therefore f(x, y), f_x(x, y)$ 在 D 上连续.

$F(x, u, v) = \int_v^u f(x, y) dy$ 关于 x 有偏导数, 且

$$\frac{\partial F}{\partial x} = \int_v^u \frac{\partial}{\partial x} f(x, y) dy, \text{ 连续.}$$

$$\frac{\partial F}{\partial u} = f(x, u); \quad \frac{\partial F}{\partial v} = -f(x, v) \text{ 连续.}$$

$\therefore F$ 有连续偏导数 $\Rightarrow F(x, u, v)$ 连续可微

$\therefore \varphi(x), \psi(x) \in C^1[a, b]$,

$$\therefore J(x) = F(x, \psi(x), \varphi(x)) \in C^1[a, b]$$

$$\text{A } J'(x) = F_x + F_u \cdot \psi'(x) + F_v \cdot \varphi'(x)$$

$$= \int_{\varphi(x)}^{\psi(x)} \frac{\partial}{\partial x} f(x, y) dy + f(x, \psi(x)) \cdot \psi'(x) - f(x, \varphi(x)) \cdot \varphi'(x).$$

定理5. (积分次序的可交换性).

设 $D = [a, b] \times [\alpha, \beta]$. $f(x, y) \in C(D)$. 则

$$\int_a^b \left(\int_{\alpha}^{\beta} f(x, y) dy \right) dx = \int_{\alpha}^{\beta} \left(\int_a^b f(x, y) dx \right) dy$$

证明: $I(x) \triangleq \int_{\alpha}^{\beta} f(x, y) dy \in C[a, b]$.

$F(x) \triangleq \int_a^x I(x) dx \in C^1[a, b]$ 且

$$F'(x) = I(x) = \int_{\alpha}^{\beta} f(x, y) dy$$

$$\int_{\alpha}^{\beta} \left(\int_a^x f(x, y) dx \right) dy \triangleq \int_{\alpha}^{\beta} g(x, y) dy: \text{ 关于 } x \text{ 可积, 且}$$

$\frac{\partial}{\partial x}(x, y) = 1$ ~~且~~ $g(x, y) \triangleq \int_a^x f(x, y) dx \in C(D)$.

$$\frac{\partial g}{\partial x} = f(x, y) \in C(D).$$

$$\begin{aligned} \frac{d}{dx} \int_{\alpha}^{\beta} \left(\int_a^x f(x, y) dx \right) dy &= \int_{\alpha}^{\beta} \frac{\partial}{\partial x} (g(x, y)) dy \\ &= \int_{\alpha}^{\beta} f(x, y) dy = I(x) \end{aligned}$$

$$= F'(x)$$

$$\therefore \int_{\alpha}^{\beta} \left(\int_a^x f(x, y) dx \right) dy = F(x) + C \quad \forall x \in [a, b].$$

C 是常数

$$\text{If } z=a: \quad t_2=0 = t_2 = F(a) + C. = C$$

$$F(z) \triangleq \int_a^z I(x) dx: \quad F(a)=0 \Rightarrow C=0.$$

$$\therefore F(z) = \int_a^\beta \left(\int_a^z f(x,y) dx \right) dy = \int_a^z I(x) dx.$$

$$\text{If } z=b: \quad \int_a^\beta \left(\int_a^b f(x,y) dx \right) dy = \int_a^b I(x) dx \\ = \int_a^b \left(\int_a^\beta f(x,y) dy \right) dx \quad \#$$

$$\text{Ex. for Fubini: } \lim_{x \rightarrow 0} \int_{-1}^1 \sqrt{x^2 + y^2} dy.$$

$$\text{If: } f(x,y) = \sqrt{x^2 + y^2}: \in C(\mathbb{R}^2).$$

$$\therefore \int_{-1}^1 \sqrt{x^2 + y^2} dy \in C(\mathbb{R}).$$

$$\lim_{x \rightarrow 0} \int_{-1}^1 \sqrt{x^2 + y^2} dy = \int_{-1}^1 \lim_{x \rightarrow 0} \sqrt{x^2 + y^2} dy \\ = \int_{-1}^1 \sqrt{y^2} dy = 2 \int_0^1 y dy = y^2 \Big|_0^1 = 1.$$

$$\text{Ex } F(y) = \int_y^{y^2} \frac{\sin xy}{x} dx. \quad \text{If } F'(y).$$

Recall: $\int_0^\infty \frac{\sin x}{x} dx = ?$

用复变函数中“留数定理”

$\int \frac{\sin x}{x} dx$: 原函数: 不能用初等函数表示.

eg: $\therefore f(x, y) = \frac{\sin xy}{x}$: 直接可求.

$\varphi(y) = y, \quad \psi(y) = y^2: \in C^1(\mathbb{R}),$

$$F(y) = \int_{\varphi(y)}^{\psi(y)} \frac{\sin xy}{x} dx = \int_{\varphi(y)}^{\psi(y)} f(x, y) dx$$

$$\therefore F'(y) = \int_{\varphi(y)}^{\psi(y)} \frac{\partial}{\partial y} f(x, y) dx + \underbrace{f(\psi(y), y)}_{\text{upper limit}} \cdot \underline{\psi'(y)} - \underbrace{f(\varphi(y), y)}_{\text{lower limit}} \cdot \underline{\varphi'(y)}$$

$$= \int_y^{y^2} \frac{\cos(xy)}{x} \cdot x dx + \frac{\sin(y^3)}{y^2} \cdot (2y) - \frac{\sin y^2}{y} \cdot 1$$

$$= \frac{1}{y} \sin(xy) \Big|_y^{y^2} + \frac{2 \sin y^3}{y} - \frac{\sin y^2}{y}$$

$$= \frac{1}{y} (\sin y^3 - \sin y^2) + \frac{2 \sin y^3}{y} - \frac{\sin y^2}{y}$$

$$= \frac{3 \sin y^3 - 2 \sin y^2}{y}$$

作业. P.169. 2, 4, 5, 6.